

- 4 The photograph shows an aquatic plant, *Cabomba aquatica*.



(Source: © Dorling Kindersley Ltd / Alamy Stock Photo)

This plant grows in water in Asia.

A student observed that the shortest plants were always seen in water shaded by overhanging trees.

The student thought that reduced photosynthesis may be limiting the growth of these plants.

The student formed the following hypothesis:

Low light intensity reduces the rate of photosynthesis in Cabomba aquatica.

Plan an investigation to find evidence to support or reject this hypothesis.

Your answer should give details under the following headings.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(2)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(b) Devise a detailed method, including how you would control and monitor important variables.

(9)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the entire width of the page. There is no handwriting or other markings on the paper.

(d) Suggest **two** limitations of your proposed method.

(2)

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(Total for Question 4 = 16 marks)

TOTAL FOR PAPER = 50 MARKS